

TrES-3b

R-0112

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240405 · created 2026-07-16T04:13:28+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460405.9676 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.02
Transit depth (Rp/Rs)^2	3.69 +/- 0.77 [%]
Orbital Inclination (inc)	81.68 +/- 0.77
Airmass coefficient 1 (a1)	0.959 +/- 0.01
Airmass coefficient 2 (a2)	0.035 +/- 0.034
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#2 - [292, 81]
Optimal Aperture	3.94
Optimal Annulus	5.45
Transit Duration (day)	0.0564 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.02 vs 0.16309 (deviation 1.45 σ)
Duration fit vs published	0.0564 d vs 0.0567 d (deviation 0.03 σ)
Mid-time O-C	-4.6 min
Depth SNR	9.6
Residual scatter	1.07 %

Light curve

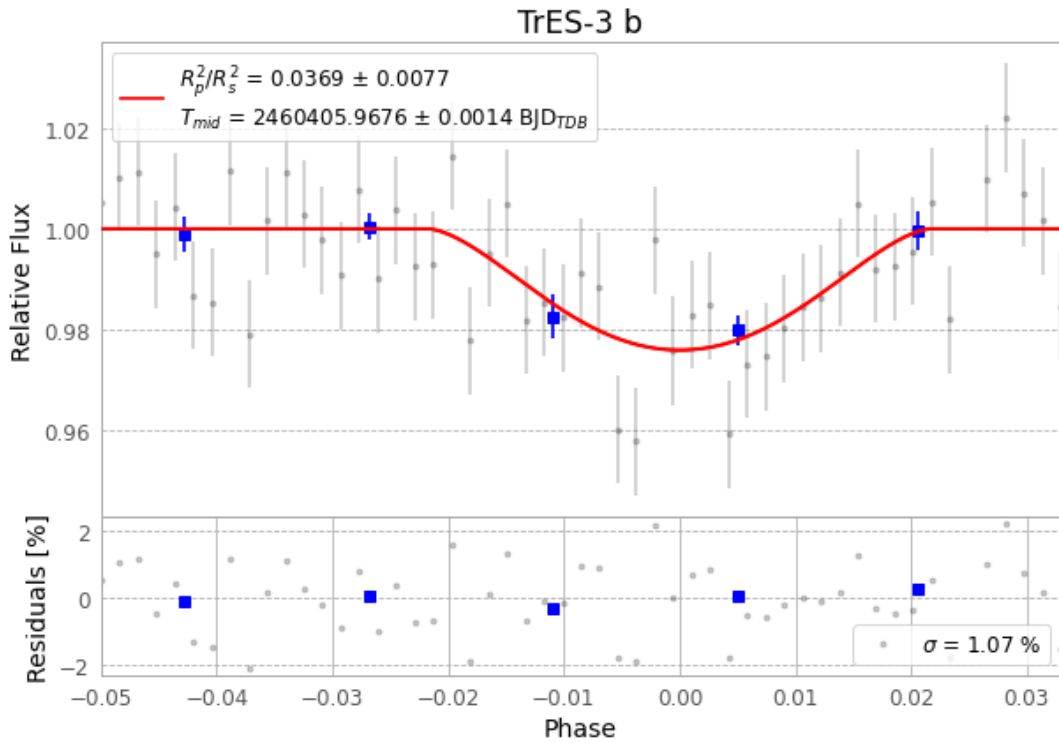


Figure 1 — FinalLightCurve_TrES-3 b_2024-04-05.png (SHA-256 75592ccb6c172ab7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c5cdccb5251d39bf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

53 FITS frames (35 MB), TRES-3240405093615.FITS → TRES-3240405121215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240405.log (20b249408f7f...), FinalLightCurve_TrES-3 b_2024-04-05.png (75592ccb6c17...), FinalLightCurve_TrES-3 b_2024-04-05.csv (d152fb4bb843...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 04:13Z.